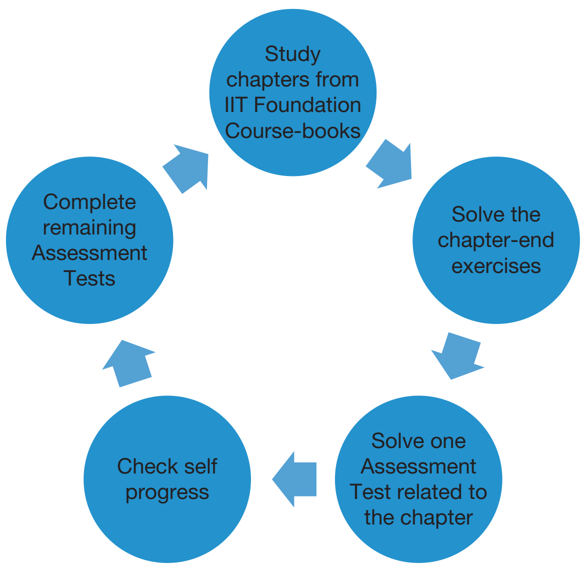
How to Use the Practice Book

Many times, students face significant challenges in answering application level questions in Physics, Chemistry and Mathematics. These Practice Books will enhance their problem-solving skill which will definitely lead to a strong subject foundation. The entire practice book series are recommended to be used alongside IIT Foundation course-books.

Students can refer the following steps while using the practice books:



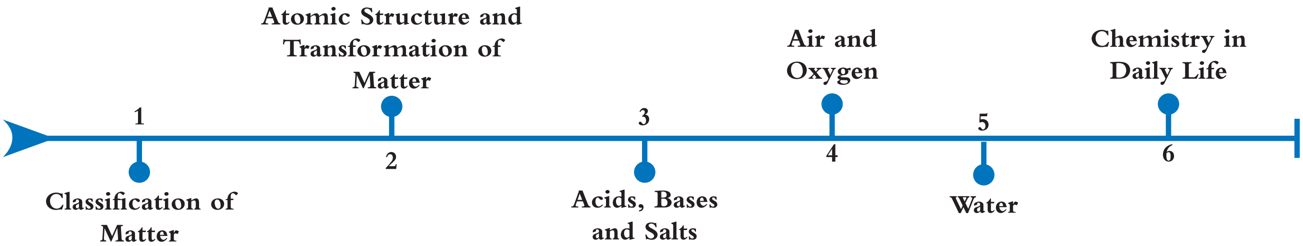
While preparing for Foundation courses, students need to learn the fundamental concepts with utmost clarity. In order to successfully complete the IIT Foundation course, one must prepare profoundly. Consistent hard work, practice and perseverance are needed throughout the year.

During any competitive examination, one must exercise clinical precision with speed since the average time available to respond to a question is hardly a minute. The aspirants should be conceptually excellent in the subject owing to the negative marking in the examination. A better practice to solve the paper would be to go for the easiest questions first and then gradually progress to the more complicated ones.

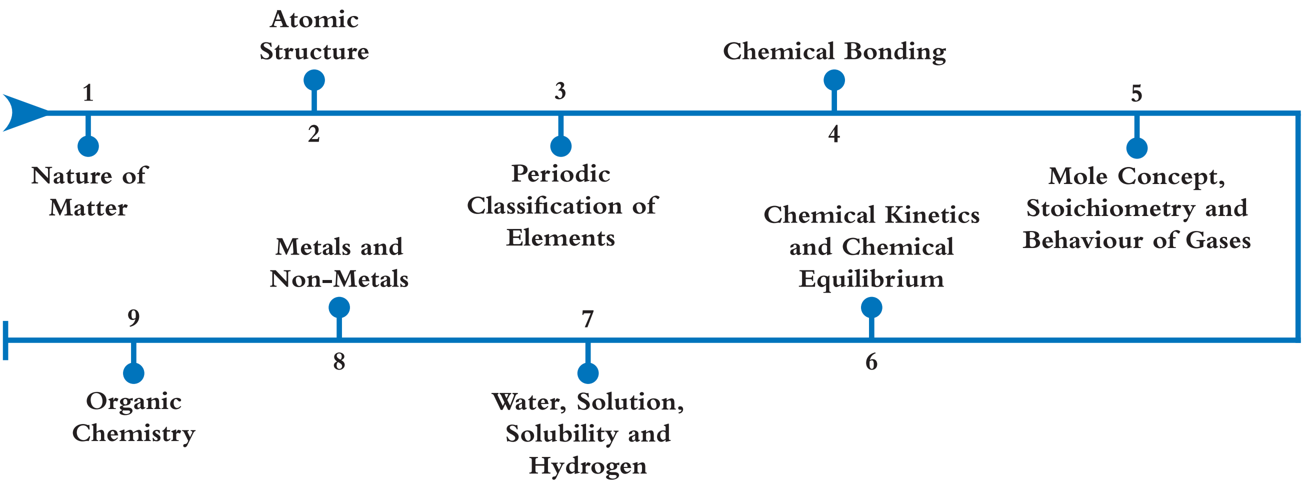
Regular practice of MCQs will assist the aspirants in preparing for the examination. In a nutshell, hard work, conceptual clarity and self-assessment are the essential ingredients to achieve success in competitive examinations. IIT Foundation course-books play an important role in understanding the concepts. Student need to read-up on all concepts/theories in a regular and systematic manner.

Course-book Chapter Flow

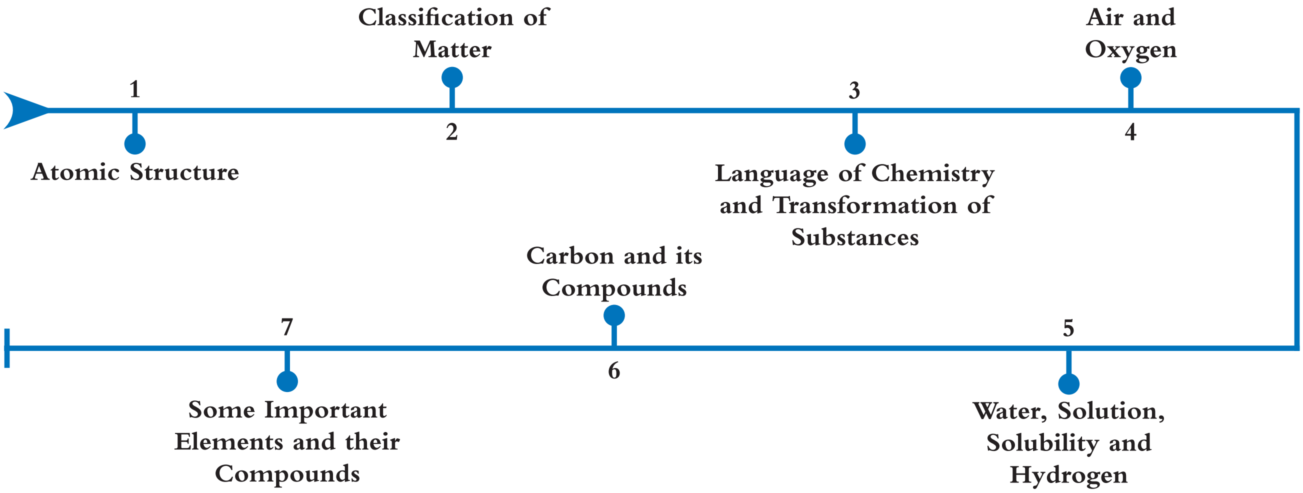
Class 7



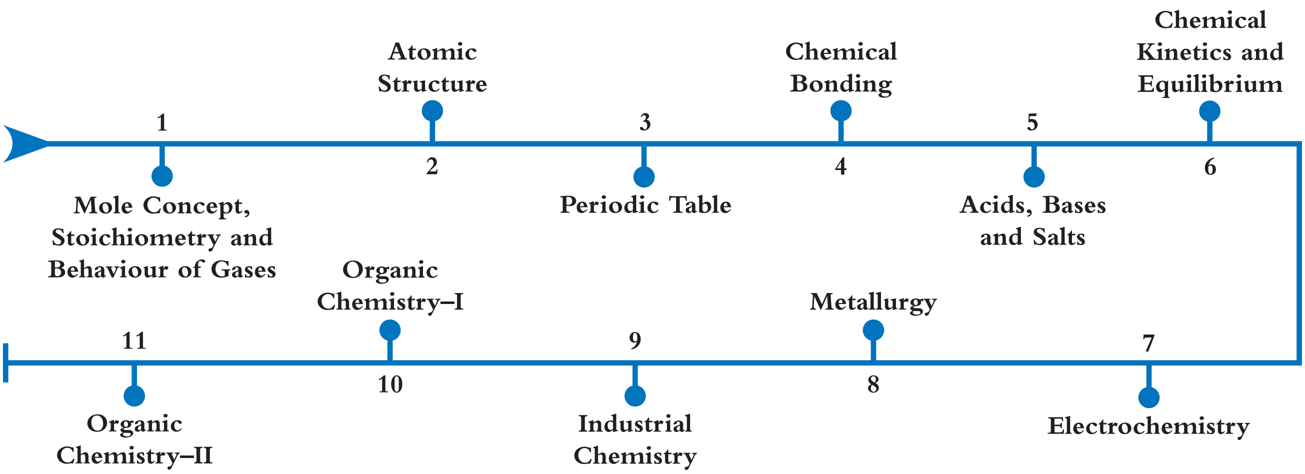
Class 9



Class 8



Class 10



Atomic Structure 1

Reference: *Coursebook - IIT Foundation Chemistry Class 8; Chapter - Atomic Structure; Page number - 1.1–1.14*

Assessment Test I

Time: 30 min.

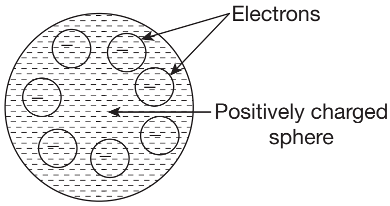
Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Which of the following is a contradiction to Dalton's atomic theory?

(a) Discovery of isotopes
(b) Law of definite proportions
(c) Law of conservation of mass
(d) Discovery of nucleus

2. An atomic model is represented by the figure given below.



Identify the fact (from following statements) which could be successfully explained by the given atomic model.

(a) Total positive charge is equal to total negative charge.
(b) Positively charged particles are shielded from negatively charged particles.
(c) The existence of nucleus in an atom.
(d) Electrons are universal constituents of matter.

3. Identify the sets of orbits (electronic transitions) associated with release of energy.

(A) $L \rightarrow M$ (B) $K \rightarrow M$ (C) $L \rightarrow K$
(D) $M \rightarrow L$ (E) $M \rightarrow N$ (F) $N \rightarrow K$

(a) ACD (b) BCE (c) CDF (d) CDE

4. **Assertion (A):** The orbits in which electrons are present are called stationary orbits.

Reason (R): Electrons remain stationary in those orbits.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

5. An atom 'X' has a mass of protons 36,740 times that of electrons. Identify the correct representation of the atom. The number of neutrons is 1 unit less than the protons.
(a) ${}_{10}X^{20}$ (b) ${}_{10}X^{22}$ (c) ${}_{20}X^{40}$ (d) ${}_{20}X^{39}$

6. If 'a' is the number of electrons in a dipositive ion and it has three neutrons more than the number of protons, identify the atomic number and mass number.
(a) $a + 2, a + 5$ (b) $a + 2, 2a + 7$
(c) $a + 5, 2a + 7$ (d) $a + 5, 2a + 5$

7. Which of the following corresponds to the incorrect match with respect to their formation?
(A) O^{-2} ion – Loss of two electrons (B) N^{-3} ion – Gain of three electrons
(C) Mg^{+2} ion – Loss of two electrons (D) K^{+} ion – Gain of one electron
(a) A and B (b) B and C
(c) A and D (d) C and D

8. Correct the given statements by replacing the underlined words and choose the correct option.
(i) Protons and neutrons together correspond to <u>subatomic particles</u>.
(ii) Rutherford's theory proposed the concept of <u>stationary orbits</u>.
(iii) Isotopes differ in the number of <u>protons</u>.
(iv) Isobars possess the same number of <u>neutrons</u>.

(i)	(ii)	(iii)	(iv)
(a) Nucleons	Protons	Neutrons	Electrons
(b) Mass number	Nucleus	Neutrons	Nucleons
(c) Atomic number	Protons	Electrons	Electrons
(d) Nucleons	Nucleus	Electrons	Protons

9. **Assertion (A):** In α-ray scattering experiment, most of the α particles were completely rebounded.
Reason (R): The entire positive charge is concentrated in the small central part known as nucleus.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

Space for rough work

10. Arrange the important points of various theories in chronological order.

(A) Presence of electrons in K, L, M and N shells.

(B) Combination of atoms in simple numerical ratio.

(C) Electrical neutrality of atom.

(D) The diameter of nucleus is 10^{-5} times that of the atom.

(a) CBAD (b) DCBA (c) BCDA (d) BDAC

11. Arrange the following pairs in the order of metals, non-metals, noble gases, isotopes and isobars:

(A) ${}_{Z}A^{X}, {}_{Z+2}B^{X}$ (B) ${}_{10}X^{20}, {}_{18}Y^{36}$

(C) ${}_{Z}A^{X}, {}_{Z}A^{X+2}$ (D) ${}_{8}X^{16}, {}_{16}Y^{32}$

(E) ${}_{12}A^{24}, {}_{20}Y^{40}$

(a) EDBCA (b) DEBAC

(c) DBECA (d) EBDAC

12. Match the values of Column A with those of Column B.

Column A (Atomic Number)	Column B (Ratio of Electrons in Penultimate Shell to Valence Shell)
(i) 21	(A) 8:1
(ii) 30	(B) 4:1
(iii) 20	(C) 9:2
(iv) 19	(D) 1:1
(v) 18	(E) 9:1

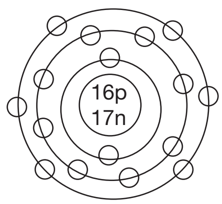
(a) (i) → (C); (ii) → (E); (iii) → (B); (iv) → (A); (v) → (D)

(b) (i) → (B); (ii) → (D); (iii) → (E); (iv) → (A); (v) → (C)

(c) (i) → (E); (ii) → (B); (iii) → (C); (iv) → (D); (v) → (A)

(d) (i) → (C); (ii) → (E); (iii) → (A); (iv) → (B); (v) → (D)

13. The geometric representation of a certain species is given below. On the basis of the representation, identify the wrong statement.



(a) After its atomic number increases by 3 units, the valence shell becomes the penultimate shell.

(b) The outermost shell can accommodate 12 more electrons.

(c) The anion formed from it has the configuration of neon.

(d) It is the geometric representation of a neutral atom.

14. Which of the following numbers does not correspond to the maximum number of electrons in Bohr's orbit?
(a) 32 (b) 36 (c) 72 (d) 98

15. An atom of an element 'X' loses an electron to get a stable octet. 'Y' also loses two electrons to form the corresponding ion. However, the resultant ion of Y does not possess an octet. Identify the probable atomic numbers of X and Y.
(a) 19, 30 (b) 20, 29 (c) 3, 20 (d) 29, 20

Space for rough work

Assessment Test II

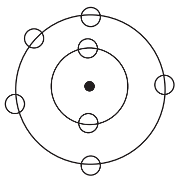
Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

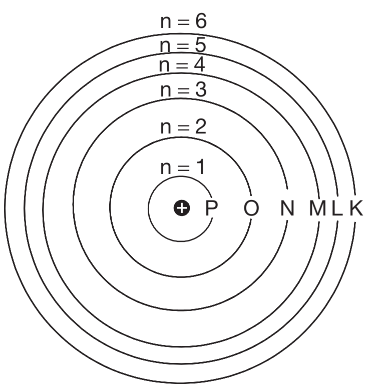
1. Identify the factor which provided the experimental basis for the postulates of Dalton's atomic theory.
   (a) Discovery of isobars
   (b) Discovery of subatomic particles
   (c) Law of definite proportions
   (d) Discovery of atomic number as the fundamental property

2. The figure given below represents an atomic model:



   Which among the following was not supported by the given atomic model?
   (a) Revolution of electrons in stationary orbits.
   (b) Relatively small size of nucleus in comparison to the size of an atom.
   (c) Increase in energy of the orbits with distance from the nucleus.
   (d) Uniform distribution of positive charge in an atom.

3. Observe the following figure and identify the one which is in contradiction to the established facts.



   (a) Distance between the orbits is not uniform.
   (b) Size of the nucleus is not in proportion to the size of the atom.
   (c) Naming of orbits is in the reverse order.
   (d) Nucleus is at the centre.

Space for rough work

4. **Assertion (A):** Energy of orbits increases with distance from the nucleus.
Reason (R): Electrons in the first orbit do not lose energy.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

5. An atom has a mass of protons as much as 18,370 times that of an electron. The mass number is two units more than twice the atomic number. Identify the number of neutrons in the atom.
(a) 10 (b) 20 (c) 22 (d) 12

6. Assuming that x is the mass of electron and y is the total mass of protons in $_8O^{18}$, which of the following equations is correct for calculating the total mass of protons?
(a) $y = 18x - 10$ (b) $8y = \frac{x}{1837}$
(c) $\frac{y}{8} = 1837x$ (d) $y = \frac{1837x}{18}$

7. Which of the following corresponds to incorrect matches with respect to their formation?
(A) Dipositive ion $\rightarrow$ 2 electrons in valence shell in neutral atom.
(B) Unipositive ion $\rightarrow$ 7 electrons in valence shell in neutral atom.
(C) Trinegative ion $\rightarrow$ 3 electrons in valence shell in neutral atom.
(D) Dinegative ion $\rightarrow$ 6 electrons in valence shell in neutral atom.
(a) A and B (b) B and C
(c) C and D (d) B and D

8. Correct the given statements by replacing the underlined words and choose the correct option.
(i) $_{17}Cl^{35}$ has the number of neutrons equal to $\underline{35}$.
(ii) The maximum number of electrons in an orbit can be given by $\underline{n^2 + 2}$.
(iii) $\underline{_{20}Y^{39}}$ is the isobar of $_{20}X^{40}$.
(iv) The isotope of $_ZX^A$ is $\underline{_{Z+2}Y^A}$.

	(i)	(ii)	(iii)	(iv)
(a)	17	n^2	$_{18}Y^{40}$	$_xX^{A+2}$
(b)	18	n^2	$_{19}Y^{39}$	$_zX^{A+2}$
(c)	18	$2n^2$	$_{19}Y^{40}$	$_zX^{A+2}$
(d)	17	$2n^2$	$_{20}Y^{40}$	$_zX^{A+2}$

9. **Assertion (A):** Most of the α particles in α-ray scattering experiment passed straight through the gold foil.
Reason (R): The size of the nucleus is very small compared to the size of the atom.